1 July 1966

MILITARY SPECIFICATION

INDICATOR, RATE OF FUEL FLOW
SYNCHRO TYPE, MS90360

This specification is approved for use by the Naval Air Systems Command, Department of the Navy.

1 SCOPE

1.1 Scope - This specification covers design and all performance requirements for synchro type, rate of fuel flow indicators up to 5000 PPH maximum.

1.2 Classification - The indicators shall be classified as shown on MS90360.

2 APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, forms part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

| | |
|---|---|
| BB-N-411 | Nitrogen, Technical |
| DD-G-451 | Glass, Flat and Corrugated for Glazing Mirrors and other uses |
| QQ-C-320 | Chromium Plating (Electrodeposited) |
| QQ-N-290 | Nickel Plating (Electrodeposited) |
| QQ-P-416 | Plating Cadmium (Electrodeposited) |
| QQ-Z-325 | Zinc Coating, Electrodeposited Requirements for |

FSC 6620

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Military

| | |
|---|---|
| MIL-P-116 | Preservation, Methods of |
| MIL-D-1000 | Drawings, Engineering and Associated List |
| MIL-M-3171 | Magnesium Alloy, Processes for Pretreatment and Prevention of Corrosion on |
| MIL-E-5272 | Environmental Testing, Aeronautical and Associated Equipment, General Specification for |
| MIL-T-5350 | Transmitters, Synchro, Aircraft, General Specification for |
| MIL-E-5400 | Electronic Equipment, Aircraft, General Specification for |
| MIL-C-5541 | Chemicals, Films and Chemical Film Materials for Aluminum and Aluminum Alloys |
| MIL-I-6181 | Interference Control Requirements, Aircraft Equipment |
| MIL-S-6872 | Soldering Process, General Specification for |
| MIL-S-7742 | Screw Threads Standard, Optimum Selected Series, General Specification for |
| MIL-A-8625 | Anodic Coatings for Aluminum and Aluminum Alloys |
| MIL-S-20708 | Synchros, 60 and 400 Cycle, General Specification |
| MIL-P-23408 | Plating: Tin-Cadmium (Electrodeposited) |
| MIL-L-25467 | Lighting, Integral, Aircraft Instrument, General Specification for |
| MIL-C-26482 | Connectors, Electric, Circular, Miniature, Quick Disconnect |
| MIL-T-81187 | Transmitter Rate of Fuel Flow, Synchro Type |



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STANDARDS

Federal

| | |
|---|---|
| FED-STD-595 | Colors |

Military

| | |
|---|---|
| MIL-STD-130 | Identification Marking of U. S. Military Property |
| MIL-STD-143 | Specifications and Standards, Order of Precedence for the Selection of |
| MIL-STD-781 | Reliability Tests Exponential Distribution |
| MIL-STD-794 | Parts and Equipment Procedures for Packaging and Packing |
| MS24367 | Lamp, Incandescent - Miniature Integral Lighting |
| MS24519 | Lamp - Sub Miniature |
| MS33558 | Numerals and Letters Aircraft Instrument Dial Standard Form of |
| MS33565 | Pointers, Dial Standard Design of Aircraft Instrument |
| MS33566 | Meters Definition of Dissimilar |
| MS90360 | Indicator Rate of Fuel Flow Synchro Type |

DRAWINGS

U. S. Air Force

| | |
|---|---|
| 33A10-78 | Transformer Assembly - Autosyn Instrument Test Set |

2.2 Availability of documents - When requesting any of the applicable documents, refer to both title and number. Copies of this specification and applicable documents may be obtained, for a fee, from the Commanding Officer, Naval Supply Depot, Code 30, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3 REQUIREMENTS

3.1 Qualification - The indicator furnished under this specification shall be a product, samples of which have been tested, and have passed the Qualification Tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.



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3.2 Selection of Government documents - Except as provided in 3.2.1 and 3.2.2, specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.2.1 Standard parts - With the exception of 3.2.2, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.2.2 Commercial parts - Commercial parts having suitable properties shall be used, when on the date of invitations for bids, there are no suitable standard parts. In any case, commercial parts, such as screws, bolts, nuts, and cotter pins, having suitable properties, may be used provided:

(a) They can be replaced by the standard parts (MS or AN) without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and if practicable, on the contractor's drawings.

3.3 Materials - Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.3.1 Critical materials - Noncritical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.3.2 Nonmagnetic materials - Nonmagnetic materials shall be used for all parts of the indicator except where magnetic materials are essential.

3.3.3 Metals - Metals shall be of the corrosion resistant type or shall be suitably protected as specified herein to resist corrosion due to fuels, salt spray, or atmospheric conditions to which the indicator may be subjected when in storage or during normal service life.

3.3.3.1 Dissimilar metals - Dissimilar metals as defined in MS33586, except those in hermetically sealed atmosphere as described in Filling medium, shall not be used in intimate contact with each other unless protection against electrolytic corrosion is provided.

3.3.3.2 Magnesium alloy parts - Magnesium alloy parts shall be treated in accordance with MIL-M-3171. Where abrasion resistance is a factor, an anodic treatment approved by the procuring activity shall be used.



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3.3.3.3 Aluminum alloy parts - Unless otherwise specified, aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625. Small holes, pipe threads, and case inserts need not be anodized. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.3.3.4 Iron and steel parts - Iron and steel parts which are in hermetically sealed cases shall be cadmium plated in accordance with QQ-P-416. Parts which are not in sealed cases shall be chromium, nickel, or zinc plated in accordance with QQ-C-320, QQ-N-290, or QQ-Z-325 respectively. Parts in a confined space in the presence of organic material shall be tin-cadmium plated in accordance with MIL-P-23408. The class and type of plating shall meet the requirements of 3.3.4.

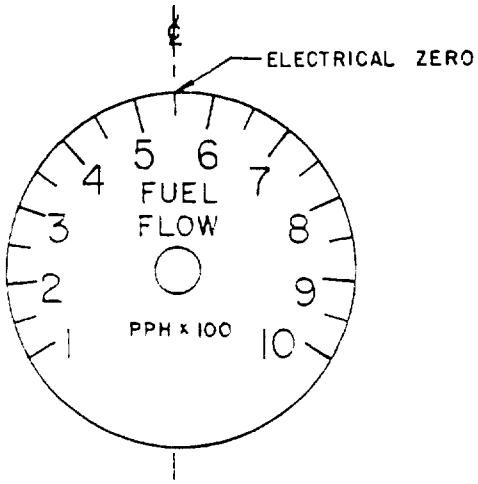
3.3.4 Protective treatment - When materials are used in the construction of the indicator that are subject to atmospheric or environmental conditions likely to cause corrosion in normal service life, they shall be protected against corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. Finishes and protective coatings which will crack, chip, or scale during normal service life or are affected by extremes of atmospheric or environmental conditions shall not be used.

3.3.5 Fumes and vapors - Materials used in the construction of the indicators shall not produce corrosive, deleterious, or toxic fumes or vapors under the conditions specified herein.

3.4 Design and construction - The outline dimensions of the indicator shall conform to the requirements of MS90360. The indicator shall provide remote indication of rate of fuel flow in pounds per hour (PPH) when connected to a transmitter conforming to MIL-T-81187. The calibration of fuel flow versus angular degrees shall be as shown in Figure 1. Intermediate graduations shall be linear within ±0° 15' of arc. Electrical zero shall be as defined in 6.3. The indicator shall be so constructed that no parts will work loose in service. Any synchro receiver used shall have electrical characteristics compatible to those defined in MIL-S-20708. The [illegible] shall withstand the stresses and vibrations and other conditions incident to shipping, storage, installation, and service. Pivots, bearings, and gears shall neither bind nor shake and shall be as free frictionless as practicable.



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Note: (1) Example shown MS90360-2 (100-1000 PPH) Scale
(2) Angles Tolerances ± 0° 15'

| Marking | Height or Length Inch ± 0.016 | | Width of Line or Graduation Inch ± 0.005 | | Finish |
|---|---|---|---|---|---|
| | 2 Inch | 3 Inch | 2 Inch | 3 Inch | |
| Scale Numerals (See Note) | 0.188 | 0.250 | 0.031 | 0.031 | Lusterless White Color No. 37875 FED-STD-595 |
| Major Graduations | 0.156 | 0.219 | 0.031 | 0.031 | |
| Minor Graduations | 0.125 | 0.156 | 0.020 | 0.020 | |
| Other Graduations, if any | 0.062 | 0.078 | 0.016 | 0.016 | |
| Electrical Zero Mark | 0.031 | 0.031 | 0.010 | 0.010 | |
| Lettering "Fuel Flow" | 0.125 | 0.188 | 0.020 | 0.031 | |
| Lettering "PPH x 100" | 0.094 | 0.125 | 0.016 | 0.020 | |

Figure 1. Dial Markings



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Figure 4 (Cont'd) Calibration Tables

Clockwise From Point Opposite Electrical Zero Mark

MS90360 (Dash -1, -2, -3, -4, -5 and -9)

| -1 | -2 | -5 | -9 | | | -3 | | | -4 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| PPR | PPR | PPR | PPR | Degrees | | PPR | Degrees | | PPR | Degrees | |
| 50 | *100 | *20 | *40 | 5 | 30 | 120 | 5 | 30 | 150 | 5 | 30 |
| 75 | 150 | 30 | 60 | 7 | 00 | 150 | 6 | 15 | *200 | 6 | 40 |
| *100 | *200 | *40 | *80 | 9 | 30 | *200 | 7 | 30 | 250 | 8 | 00 |
| 125 | 250 | 50 | 100 | 11 | 00 | 250 | 8 | 45 | 300 | 9 | 30 |
| 150 | *300 | *60 | 120 | 11 | 30 | *300 | 9 | 00 | 350 | 10 | 00 |
| 175 | 350 | 70 | 1400 | 12 | 00 | 350 | 10 | 15 | *400 | 9 | 30 |
| *200 | *400 | *80 | *160 | 13 | 30 | *400 | 12 | 30 | 450 | 9 | 00 |
| 225 | 450 | 90 | 180 | 15 | 00 | 450 | 13 | 45 | 500 | 10 | 30 |
| 250 | *500 | *100 | 200 | 16 | 30 | *500 | 14 | 00 | 550 | 11 | 00 |
| 275 | 550 | 110 | 220 | 18 | 00 | 550 | 15 | 15 | *600 | 12 | 30 |
| *300 | *600 | *120 | *240 | 19 | 30 | *600 | 16 | 30 | 650 | 13 | 00 |
| 325 | 65 | 130 | 260 | 21 | 00 | 65 | 17 | 45 | 700 | 13 | 30 |
| 350 | *700 | *140 | 28 | 22 | 30 | *700 | 19 | 00 | 750 | 14 | 00 |
| 375 | 750 | 150 | 300 | 24 | 00 | 750 | 20 | 15 | *800 | 15 | 30 |
| *400 | *800 | *160 | *320 | 24 | 30 | *800 | 21 | 30 | 850 | 15 | 00 |
| 425 | 850 | 170 | 340 | 25 | 00 | 850 | 22 | 45 | 900 | 16 | 30 |
| 450 | *900 | *180 | 360 | 27 | 30 | *900 | 24 | 00 | 950 | 17 | 00 |
| 475 | 950 | 190 | 380 | 28 | 00 | 950 | 24 | 15 | *1000 | 18 | 30 |
| *500 | *1000 | *200 | *400 | 30 | 30 | *1000 | 25 | 30 | 1050 | 19 | 00 |
| | | | | | | 1050 | 26 | 45 | 1100 | 19 | 30 |
| | | | | | | *1100 | 27 | 00 | 1150 | 20 | 00 |
| | | | | | | 1150 | 28 | 15 | *1200 | 21 | 30 |
| | | | | | | *1200 | 30 | 30 | 1250 | 21 | 00 |
| | | | | | | | | | 1300 | 22 | 30 |
| | | | | | | | | | 1350 | 23 | 00 |
| | | | | | | | | | *1400 | 24 | |
| | | | | | | | | | 1450 | 24 | 00 |
| | | | | | | | | | 1500 | 25 | 30 |
| | | | | | | | | | 1550 | 26 | 00 |
| | | | | | | | | | *1600 | 27 | 30 |
| | | | | | | | | | 1650 | 27 | 00 |
| | | | | | | | | | 1700 | 28 | 30 |
| | | | | | | | | | 1750 | 29 | 00 |
| | | | | | | | | | *1800 | 0 | 00 |

NOTES '*' indicates major dial graduations and numerals to appear on the scale. If the first numeral of the column is not marked '*' it shall appear on the scale with a calibration mark and the number shall be approximately one half the height of the other major calibration numbers. A — indicates a zero graduation mark to appear on the dial without numerals.



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Figure 1 (Cont'd) Calibration Tables

Clockwise From Point Opposite Electrical Zero Mark

MS90360 ( Dash 6, 7, 8, 10 and 11)

| -6 -11 | | | -7 | | -8 | | -10 | |
|---|---|---|---|---|---|---|---|---|
| PPH | PPH | Degrees | PPH | Degrees | PPH | Degrees | PPH | Degrees |
| 250 | 500 | 58° 30' | 300 | 58° 30 | 350 | 58° 30' | 450 | 58° 30' |
| 300 | 600 | 63° 54' | 400 | 67° 30' | *500 | 70° 08' | 600 | 67° 30' |
| 400 | 800 | 74° 42 | *500 | 78 30 | 750 | 80° 30 | 800 | 79° 37' |
| *500 | *1000 | 85° 30 | 600 | 85° 30 | *1000 | 90° 52' | *1000 | 91° 30 |
| 600 | 1200 | 96° 18 | 700 | 94° 30 | 1250 | 128° 14' | 1200 | 102° 30' |
| 700 | 1400 | 107° 06' | 800 | 102° 30 | *1500 | 147° 36' | 1400 | 113° 30' |
| 800 | 1600 | 117° 54' | 900 | 112° 30' | 1750 | 166° 50' | — | |
| 900 | 1800 | 128° 42' | *1000 | 121° 30' | *2000 | 184° 20' | 1600 | 124° 30 |
| *1000 | *2000 | 139° 30' | 1100 | 130° 30 | 2250 | 200° 42 | 1800 | 136° 30 |
| 1100 | 2200 | 150° 18' | 1200 | 139 30 | *2500 | 215° 04 | *2000 | 151° 30 |
| 1200 | 2400 | 161° 06 | 1300 | 148 30 | 2750 | 254° 26' | 2200 | 162° 30 |
| 1300 | 2600 | 171° 54' | 1400 | 157° 30' | *3000 | 270° 48' | 2400 | 175° 30 |
| 1400 | 2800 | 182° 42 | *1500 | 166° 30 | 3250 | 282° 10 | — | |
| *1500 | *3000 | 193° 30' | 1600 | 175° 30' | *3500 | 301° 30 | 2600 | 187° 30 |
| 1600 | 3200 | 204° 18' | 1700 | 184° 30' | | | 2800 | 199° 30 |
| 1700 | 3400 | 215° 06' | 1800 | 193° 30' | | | *3000 | 211 30 |
| 1800 | 3600 | 225° 54' | 1900 | 202° 30' | | | 3200 | 222° 30 |
| 1900 | 3800 | 236° 42' | *2000 | 211° 30' | | | 3400 | 233° 30 |
| *2000 | *4000 | 247° 30' | 2100 | 220° 30' | | | — | |
| 2100 | 4200 | 258° 18' | 2200 | 229 30 | | | 3600 | 247° 30 |
| 2200 | 4400 | 269° 06' | 2300 | 238° 30' | | | 3800 | 258° 30 |
| 2300 | 4600 | 279° 54 | 2400 | 247° 30' | | | *4000 | 271° 30 |
| 2400 | 4800 | 290° 42' | *2500 | 256° 30 | | | 4200 | 282° 30 |
| *2500 | *5000 | 301° 30' | 2600 | 265° 30' | | | 4400 | 293° 30 |
| | | | 2700 | 274° 30' | | | **4500 | 70.° 30 |
| | | | 2800 | 283° 30' | | | | |
| | | | 2900 | 292 30 | | | **4500 to appear in same size numerals as 450 | |
| | | | *3000 | 301° 30 | | | | |

NOTES "*" indicates major dial graduations and numerals to appear on the scale If the first numeral of the column is not marked "*" it shall appear on the scale with a calibration mark and the number shall be approximately one half the height of the other major calibration numbers A — indicates a minor graduation mark to appear on the dial without numerals



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3.4.1 Integral lighting - The lighted type indicator shall be provided with integral lighting in accordance with MIL-L-25467.

3.4.1.1 Lamps - The lamps shall be located within the indicator enclosure. Lamps conforming to MS24367 or MS24515 may be used in lieu of, or in addition to, the lamps specified in MIL-L-25467.

3.4.1.2 Maintenance - The design shall be such as to facilitate as much as possible disassembly, repair or overhaul, service maintenance, and reassembly using those tools and items of maintenance equipment which are normally available as commercial standards.

3.4.2 Case - The case shall conform to MS90300. The case shall be finished with a lusterless black material, Color No. 37038 of FED-STD-595. The finishing material shall be a durable type to withstand usage encountered in service.

4.2.1 Hermetic sealing - The case shall provide a hermetically sealed enclosure for all of the mechanism. The case shall be so constructed that it may be opened, the mechanism removed, replaced, and the case re-sealed at least three times. This shall be possible without the use of any special tool, jig or fixture, unless such device is specifically approved by the procuring activity. The sealing of the case shall not be dependent upon any material which will be adversely affected by any environment to which the indicators may be subjected in normal use in military aircraft.

3.4.2.2 Filling medium - The filling medium shall be a mixture of 90 percent nitrogen and ten percent helium. The nitrogen used shall be in accordance with BB-N-411, Type I, Class I, Grade C. The filling medium shall contain not more than 0.00r milligram of water vapor per liter (dew point -73° C) at the filling pressure. The absolute pressure of the filling medium in the case shall be 1 to 1 atmosphere.

3.4.3 Weight - The weight of the completed assembled single indicator shall not exceed 0.7 pound, the dual indicator weight shall not exceed one pound.

3.4.4 Screw threads - Screw threads 0.060 inch or diameter or larger shall be in accordance with MIL-S-7742.

3.5 Electrical requirements -

3.5.1 Input power requirements - Exclusive of lighting power, the indicator shall operate on AC power (single phase), 26V ±10%, 400 cps ±5%, 3.5 watt max. (For dual indicators, 4.2 watt max.)

3.5.2 Undervoltage protection - The indicator shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage is within the specified limits.

3.5.3. Electronic parts - Electronic parts and the application thereof shall be in accordance with MIL-E-5400. Parts that do not appear



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on approved lists in accordance with MIL-E-5400 shall not be used unless approved by the procuring activity.

3.5.4 Electric connectors - The electric connector(s) used shall conform to MIL-C-26482 and MS90360. The connector code number markings as specified in MIL-E-5400 shall be legibly and permanently applied to the case or connector of the indicator

3.5.5 Wiring - The internal and external wiring shall be as shown in Figure 2, and Figure 3 respectively Internal wiring shall be color coded, neat and accomplished in such a manner that individual wires may be easily traced The wiring shall be insulated from the instrument case except that pin J shall be connected to the case

3.6 Dial and pointer -

3.6.1 Dial - The dial shall be securely mounted in such a manner that it will not loosen or slip when the indicator is vibrated. If screws are used, they shall be so located that they will not interfere with the proper location of any dial marking.

3.6.1.1 Visibility of dial - The pointer, numerals, at least 1/2 of the shortest graduations, and all other specified markings on the dial shall be visible from any point within the frustum of a cone whose side makes an angle of 30° with a perpendicular to the dial and whose small diameter is the aperture of the case

3.6.1.2 Dial marking - The dial shall be marked as shown in Figure 1 Unless otherwise specified, all visible portions of the dial shall be lusterless black Color No. 37038 of FED-STD-595 and all markings on the dial shall be lusterless white Color No 37875 of FED-STD-595 The colors shall not change as a result of service use or subjecting the indicator to the inspections specified herein The form of the numerals and letters shall be in accordance with MS33558. Numerals shall distinctly indicate the graduation to which each applies If practicable, each numeral shall be placed so that the center of area of the numeral(s) is on the radial line joining the appropriate graduation and the center of the dial Any confusion resulting in doubt as to the graduation to which the numeral applies shall be cause for rejection. When several numerals are used in one group, the space between the numerals shall be not less than 1/64 inch

3.6.1.3 Electrical zero reference mark - The electrical zero reference mark shall appear on the dial at the angular midpoint of the graduated portion as shown on Figure 1 (See 6.4.3).

3.6.2 Pointer - The pointer(s) of the indicator shall be firmly attached to the mechanism but also shall be readily adjustable The shaded portion of the pointer(s) shall be finished lusterless white Color No 37875 of FED-STD-595 The unshaded portion shall be finished lusterless black, Color No. 37038 of FED-STD-595 The pointer shall, for single type indicators, conform to MS33585-1. For dual indicators the pointer closest to the glass shall conform to MS33585-12 and the second pointer shall



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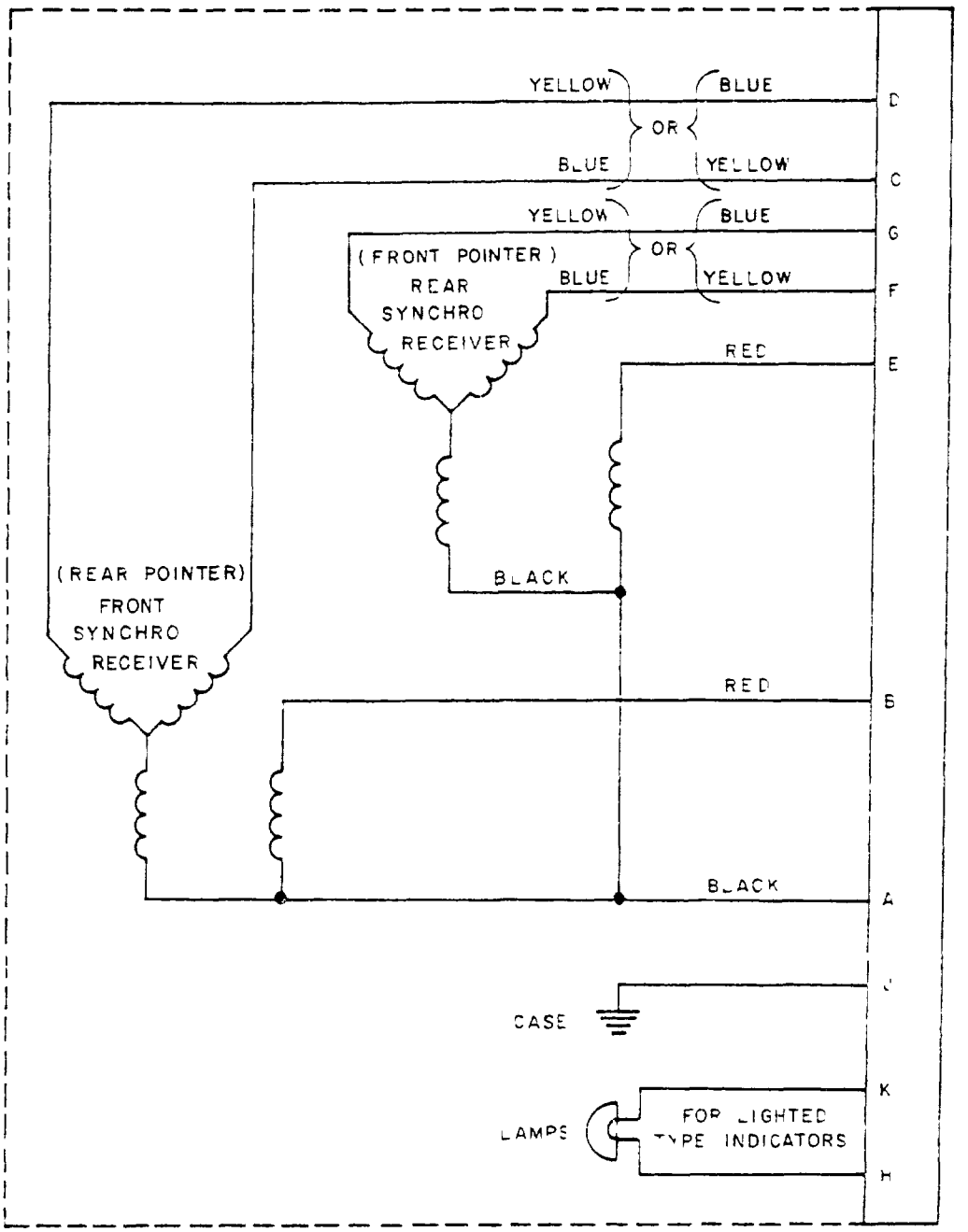
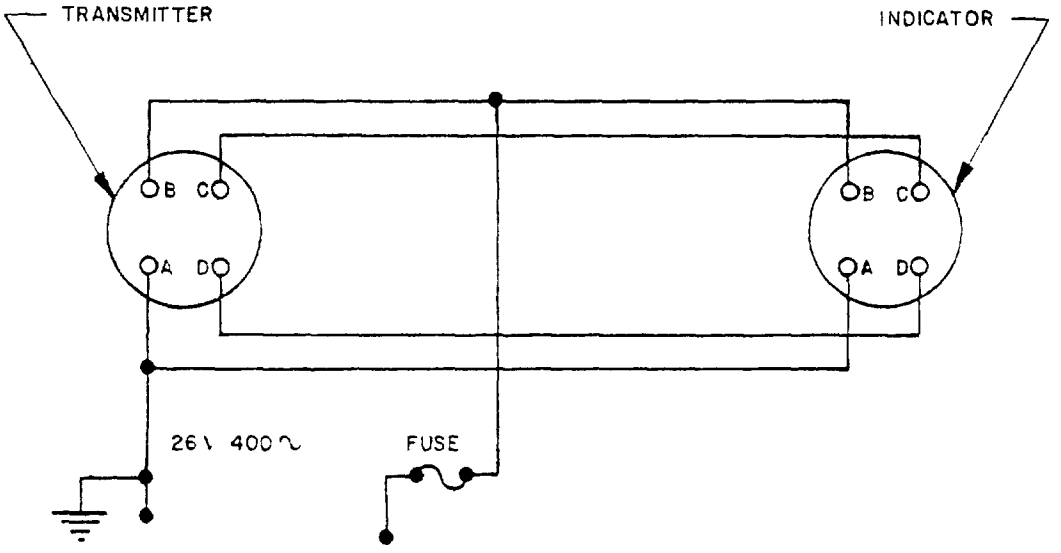
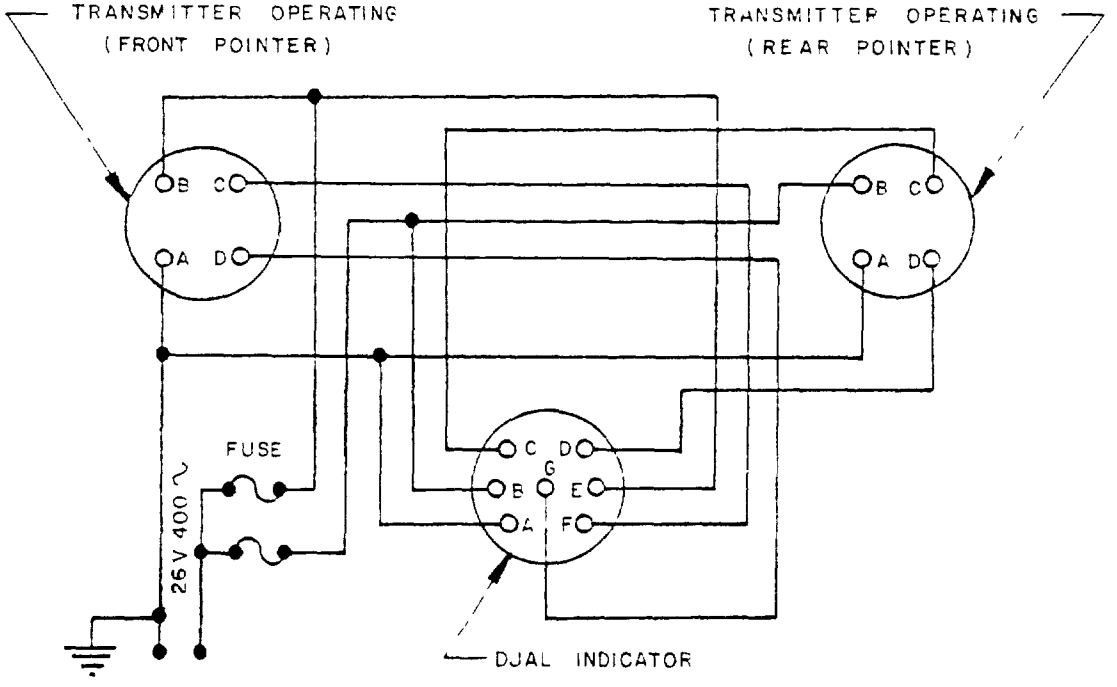



Figure 2 Internal Wiring (For Single Units, Omit Wiring to "Rear" Synchro)



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NOTE Or. indicator connectors pins J, K and H are not shown Pin J is case ground. Pins K and E are for 5V lighting power for the lighted type indicators.

Figure 3. External Wiring



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conform to MS33585-11 The pointer(s) of the lighted type indicators shall be suitable for integral lighting.

3.7 Cover glass - The quality of the cover glass shall be in accordance with DD-G-451, Type II, Quality [illegible], flaws permitted by DD-G-451 shall not interfere with reading of the dial

3.7.1 Cover glass mounting - The distance from the inner surface of the cover glass to the surface of the dial or whier the marking is applied shall not exceed 0.188 inch for single indicators or 0.270 inch for dual indicators

3.8 Performance - The indicator shall perform satisfactorily when subjected to the tests specified in Section 4 of this specification

3.8.1 Reliability -

3.8.1.1 Operational stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 1000 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use

3.8.1.2 Reliability or mean-time-between-failures - The indicator shall have 2000 hours of mean (operating) time between failures (MTBF) when tested and accepted as outlined under the requirements of 4.4.3

3.9 Identification of product -

3.9.1 Nameplate - A nameplate shall be securely attached to the exterior of the case and shall be marked in accordance with the requirements of MIL-STD-130 except that the PCI shall be omitted

3.9.2 Manufacturer's part number - The manufacturer's part number on the nameplate shall be identical with the manufacturer's engineering production drawing number including applicable dash numbers of the drawing as tabulated and covers more than one part

3.10 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-1000.

3.11 Soldering - Soldering shall be performed in accordance with MIL-S-6872

3.12 Workmanship - The indicator including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting, riveting, machine-screw assemblage, welding and brazing, and freedom of parts from burrs and sharp edges



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection - Inspection of the indicators shall be classified as follows:

(a) Qualification inspection. Qualification inspection consists of examinations and tests performed on sample indicators submitted for approval as a qualified product.

(b) Quality conformance inspection. Quality conformance inspection consists of examinations and tests performed on indicators manufactured and submitted for acceptance under contract.

4.3 Qualification inspection - The qualification inspection of the indicators shall consist of all of the examinations and tests of this specification performed in the order specified under the paragraph headed Inspection methods and shall include the reliability test for the Reliability Qualification Phase. Unless otherwise stated in the contract the tests of the Reliability Sampling Phase only are required.

4.3.1 Qualification inspection sample - Qualification inspection samples shall consist of three indicators manufactured in accordance with this specification. The indicators submitted for qualification inspection shall have been previously subjected only to the Individual inspections. The samples shall be forwarded at the contractor's expense to the laboratory designated in the Letter of Authorization.

4.3.1.1 Qualification inspection sample identification - The qualification inspection samples shall be plainly identified by durable tags, securely attached, and marked with the following information:

Sample for Qualification Inspection
INDICATOR, RATE OF FUEL FLOW ___ TO ___ PPH
SYNCHRO TYPE
Submitted by (Manufacturer's name, date)
for qualification inspection in accordance
with MIL-I-81186 under authorization
(reference letter authorizing tests)

4.4 Quality conformance inspection - The Quality conformance inspection shall consist of the Individual inspection, the Sampling plans, and the Reliability assurance tests of this specification. The contractor shall furnish all samples and shall be responsible for accomplishing all the inspections. Sampling plan B inspections shall be conducted at a



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Government laboratory designated by the procuring activity. Quality conformance inspection, except for Sampling plan B, shall be under the supervision of the Government quality control representative. The contractor shall furnish test reports showing quantitative results for all tests required by this specification signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product.

4.4.1 Individual inspection - Each indicator submitted for acceptance shall be subjected to the Individual inspection. This inspection shall determine compliance with the requirements of materials, workmanship, operational adequacy, and reliability. As a minimum, each indicator accepted shall have passed the following inspections:

Examination of product
Dielectric strength
Electrical zero
Scale error and friction
Pointer error
Illumination

4.4.2 Sampling plans - The Sampling plans shall consist of Sampling plan A and Sampling plan B inspections. The inspection samples selected for sampling tests shall first have passed the Individual inspections. The inspection samples which have been subjected to Sampling plan A inspection shall not be delivered on contract until they have been refurbished (see 6.3.5) and resubmitted and passed all the Individual inspections. Inspection samples which have been subjected to the Sampling plan B inspection shall not be delivered on contract.

4.4.2.1 Sampling plan A sample selection - Sampling plan A samples shall be selected at random in accordance with the following schedule:

| Quantity Offered for Acceptance | Quantity to be Selected for Inspection |
|---|---|
| First 25 | (See Note) |
| Next 50 | 1 |
| Next 75 | 1 |
| Next 100 | 1 |
| Each additional 200 or fraction thereof | 1 |

NOTE: When Sampling plan B is invoked, the quantity shall be zero. When Sampling plan B is to be omitted, the quantity shall be one.

When a defective indicator occurs, no items from those still on hand or later produced shall be accepted until the extent and cause of failure has been determined and appropriately corrected. In addition, when a



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failure occurs, shift to one sample out of fifteen (when Sampling plan B is omitted) and proceed as indicated.

4.4.2.1.1 Sampling plan A inspection - Each sample selected for Sampling plan A inspection shall be subjected to the following tests.

Low temperature operation
High temperature operation
Sealing
Magnetic effect
Electrical characteristics
Effect of lead length
Vibration resonance

4.4.2.2 Sampling plan B instructions - Two indicators shall be selected at random from the first 25 produced on contract and submitted within 10 days after manufacture. These samples shall be forwarded at the Contractor's expense to a Government laboratory designated by the procuring activity. Each sample shall be plainly identified by a durable tag securely attached and marked with the following information

INDICATOR, RATE OF FUEL FLOW ___ TO ___ PPH,
SYNCHRO TYPE
Submitted by (Manufacturer's name, date)
for Production Acceptance Sampling plan B
inspection, in accordance with Contract
No.
Manufacturer's part number _____

4.4.2.2.1 Sampling plan B inspection - Each sample selected for Sampling plan B inspection shall be subjected to the following tests

Sampling plan A
Vibration cycling
Low temperature exposure
High temperature exposure
Thermal shock
Fogging
Radio noise interference
Life
Internal examination
Lighting contrast
Mechanical characteristics



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4.4.3 Reliability assurance tests - Reliability assurance tests shall be conducted using MIL-STD-781. Qualification phase and sampling phase tests shall be conducted. Equipments selected for reliability assurance tests shall first have passed the indivual tests.

4.4.3.1 Reliability qualification phase - [illegible] Qualified Products Listing [illegible] a minimum of [illegible] shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the qualification phase, test level E shall be used. The accept-reject criteria for test plan I shall be used.

4.4.3.2 Reliability sampling phase tests - Samples of the equipment shall be tested as outlined in MIL-STD-781, under the section entitled "Sampling Phase of Production Reliability Tests". For the sampling phase, test level E shall be used. The accept-reject criteria for test plan II shall be used to determine the length of the tests (until an accept or reject decision is reached).

4.4.3.2.1 Lot size for sampling phase - The entire contract quantity (with the exception of those equipments used in the qualification phase) shall be considered one lot. The test program shall start the first month after the qualification phase has been completed. Samples shall be selected and placed on test in the same manner as though one month's production was a lot. The samples shall be tested until an accept or reject decision is reached. In the meantime other equipments constructed shall be shipped. The test results shall be summarized monthly for the procuring activity. The procuring activity reserves the right to stop the acceptance of equipment at any time after one or more reject decisions have been reached, pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment quality control, etc., so that the entire contract quantity will show an accept decision.

4.4.3.3 Test details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted to and approved by the procuring activity prior to the beginning of the qualification test phase of the reliability assurance tests. The following paragraphs shall be considered minimum conditions and shall apply to both the qualification and sampling phases.



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4.4.3.3.1 Duty cycle - The duty cycle shall be continuous. The duty cycle shall consist of cycling the indicator between approximately 10 percent and 80 percent of end range and return at a rate of 12 ±1 cycles per hour. The power shall be "off" 3 to 5 minutes of each hour. The indicator lighting shall be energized when the indicator is operating.

4.4.3.2.2 Performance characteristics to be measured - The Scale error and friction test, described under Inspection methods, shall be conducted at least once each week. The indicator shall be observed daily, on regular working days, for any obvious failure such as rough pointer movement

4.4.3.3.3 Failure criteria - Whenever performance characteristics fall below the acceptance requirements (4.4.3.4), at least one failure has occurred If subsequent analysis reveals that several parts have deteriorated, each should be counted a failure, unless one part caused the other parts to fail

4.4.3.3.4 Preventive maintenance - No preventive maintenance may be accomplished on the samples while they are on test

4.4.3.3.5 Disposition of samples upon completion of tests - Any sample tested may be delivered on contract provided it meets all the following requirements

(a) It is representative of production units currently being accepted

(b) It is in 'good as new' condition or has been refurbished (See 6.3)

(c) It is otherwise satisfactory

4.5 Inspection conditions -

4.5.1 Standard conditions - Unless otherwise specified, all inspection required by this specification shall be made under the following conditions:

| | |
|---|---|
| Temperature | Room ambient 25° ±5° C |
| Pressure | Normal atmospheric (approximately 29.92 inches Hg) |
| Humidity | Room ambient up to 90 percent relative humidity |
| Input Power | 26 ±0.5% AC, 400 ±5 cps<br>5 ±0.5% AC, 400 ±5 cps (lighted indicators) |

4.5.2 Test readings - Unless otherwise specified before a test reading is taken the indicator shall be vibrated using a vibrator set at 30 cps, 60 cps, or 120 cps with a maximum amplitude of 0.002 inch.



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4.5.3 Attitude - Unless otherwise specified, the indicator shall be tested in its normal operating position.

4.5.4 Electrical zero - Electrical zero shall be as defined in 6.3.3.

4.5.5 Test standards - The indicators shall be tested with a transmitting standard designed especially for the purpose. Sufficient tests shall be performed to determine that the transmitted signals provide a torque on the indicator of 2000 ±10 percent milligram millimeters per degree, and with a maximum transmitted error not over 0.20 circular degree.

4.5.6 Lighting - The indicator lighting system of lighted type indicators shall be energized and shall operate satisfactorily during all tests except thermal shock and salt spray. The indicator lighting shall operate satisfactorily after completion of each test unless otherwise specified herein.

4.5.7 Tolerances - Tolerances are given for major graduations however the same tolerances apply to all of the whole scale. (See Tables I and II.)

4.6 Inspection methods -

4.6.1 Examination of product - Each indicator shall be examined externally to determine conformance with applicable drawings and with all the requirements of this specification not covered by tests.

4.6.2 Dielectric strength - A potential of 200 volts at a commercial frequency shall be applied between each pin except pin "J" of the electrical receptacle and the case of the indicator for a minimum period of 5 seconds. There shall be no breakdown of the insulation.

4.6.3 Electrical zero - For the purpose of this test a single indicator with the glass removed shall be connected to the indicator being tested and used as the transmitter unit. The standard transmitter shall not be used. With the power source disconnected the pointer of the single indicator shall be displaced 180 degrees from the pointer position of the indicator being tested. The pointer of the single indicator shall be held in this position and power shall be applied. This procedure shall be repeated a minimum of three times after which electrical zero shall be imposed on the indicator being tested. The tip of the pointer shall be within 0.010 inch of the electrical zero reference mark, and for dual indicators within 0.010 inch of each other. (See 6.3.3.)

4.6.4 Scale error and friction - The knob of a standard transmitter shall be rotated to bring its pointer to the desired indication. Two readings shall be taken, the first before and the second after the instrument is tapped as in 4.5.2. The difference between the first and second readings shall not exceed the friction tolerance specified in Table II. The second reading shall not exceed the scale error specified in Table I. The scale readings of units of dual indicators shall be taken simultaneously. At the completion of each scale error test each unit shall be operated separately over its entire range to determine that there is no interference between units.



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TABLE I

Scale Error Tolerances
At Major Scale Graduations

| Tolerance ± Percent of End Scale Reading | |
|---|---|
| Room Temperature | Temperature Low High |
| 0.5 | 1.0 |

TABLE II

Position Error, Friction, and Vibration Tolerances

| Test | Tolerance ± Percent of End Scale Reading |
|---|---|
| Position error | 0.5 |
| Friction<br>Room temperature<br>High temperature<br>Low temperature | <br>0.5<br>1.0<br>1.0 |
| Vibration<br>Pointer oscillation<br>Pointer variation | <br>1.0<br>1.0 |



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4.6.5 Position error - The standard transmitter shall be set to approximately 50 percent of scale and locked. The reading of the indicator taken while the indicator is held in its normal operating position shall not differ from the reading while the indicator is held in any other position, by more than the amount specified in Table II

4.6.6 Illumination - After lighted type indicator shall be subjected to and meet the requirements of the following tests specified in MIL-L-3646:

(a) Operation

(b) Dial visibility

4.6.7 Low temperature operation - The indicator shall be tested in accordance with Procedure I of MIL-E-5272 after 4 hours and while still at the low temperature the indicator shall be tested for friction and scale errors. Friction shall not exceed the tolerances specified in Table II. Scale errors shall not exceed the tolerance specified in Table I. The indicator shall not require more than 3 seconds to come within the tolerance specified. The indicator shall be returned to room temperature for a minimum period of three hours and shall then be subjected to, and meet, the requirements of 4.6.4.

4.6.8 High temperature operation - The indicator shall be tested in accordance with Procedure II of MIL-E-5272 except that the duration of the high temperature shall not exceed 4 hours. After 4 hours and while still at the high temperature the indicator shall be tested for friction and scale errors. Friction shall not exceed the tolerances specified in Table II. Scale errors shall not exceed the tolerances specified in Table I. The indicator shall be returned to room temperature for a minimum period of 3 hours and shall then be subjected to, and meet, the requirements of 4.6.4.

4.6.9 Sealing - The indicator shall be immersed in deaerated distilled water. The distilled water shall be treated by adding approximately 2 percent by weight of a wetting agent, the alkyl-aryl-sodium-sulfonate type (40 percent active ingredient). The absolute pressure of the air above the water shall be reduced to approximately 5 inches Hg. Any bubbles coming from within the instrument case shall be considered as leakage and shall be cause for rejection. Bubbles which are the result of entrapped air or the gaskets exterior surfaces of the case shall not be considered as a leak. Helium leak detector or other methods of test equal or superior in sensitivity to the immersion test method described above may be used upon approval by the procuring activity. When a leak detector is employed for conducting this test, the leak rate shall not exceed 1 micron cubic foot per hour at a pressure differential of approximately 1 atmosphere.

4.6.10 Magnetic effect - The indicator shall be properly connected and power applied. The indicator shall be revolved about a short bar-magnet



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compass with the nearest part of the indicator 5-1/2 inches from the bar-magnet. The compass shall have its compensating magnets removed and shall be set up in a uniform magnetic field whose horizontal intensity is between 0.17 and 0.19 oersted. The indicator shall be revolved in a horizontal plane which is perpendicular to the axis of the bar-magnet. The indicator shall be held in positions 0, 45, 90, 135, 180, 225, 270, and 315 degrees from the initial position. At each of these positions the indicator shall be rotated 360 degrees about its horizontal axis. Deflection of the compass shall not exceed 1 degree. The test shall be repeated, without power and the same tolerance shall apply.

4.6.11 Electrical characteristics - The indicator synchro shall be tested for electrical characteristics with normal excitation of 26 volts, 400 cycles, applied to the primary (rotor). The current drawn by the indicator shall not exceed 120 milliamperes, with secondaries (stator) open circuited. The power consumption of the indicator (either synchro in dual indicators) shall not exceed 0.60 watt. The maximum open circuit voltage measured between any two secondary leads shall be 11.8 ±0.3 volts rms. The variation between the three minimum secondary voltages of any synchro shall not exceed 0.2 volt. These values do not include lighting power.

4.6.12 Effect of lead length - The indication shall not change more than 1 degree of arc when the lead length between indicator and transmitter is changed from 5 to 500 feet of No. 20 copper wire. During this test the power ground wire, Pin A, shall be grounded at both the indicator and transmitter to simulate aircraft ground.

4.6.13 Vibration resonance - The indicator shall be tested in accordance with Procedure XII, curve E of MIL-E-5272. The indicator shall be operating and indicating at one of three points chosen by the inspector. These points shall be approximately equidistant, numerically, on the scale and the tests shall be conducted at each point. Pointer oscillation and pointer variation shall not exceed that specified in Table II during the resonance test.

4.6.14 Vibration cycling - The indicator shall be tested in accordance with Procedure XII, curve B of MIL-E-5272. It shall be operating and indicating approximately 60 percent of end scale. The scale and friction errors at the completion of the vibration cycling shall not exceed the tolerance allowed at Room temperature (Tables I and II). No screws or other parts shall become loosened or damaged as a result of the vibration.

4.6.15 Low temperature exposure - The indicator shall be tested in accordance with Procedure II of MIL-E-5272. The indicator shall be operated near the end of the first 72 hour period to determine that the pointer moves smoothly over the entire scale range. The indicator shall also be tested near the end of the final 24 hour period by subjecting it to the scale error and friction test except that in every case the indicator shall not require more than 4 seconds to come within the tolerances specified. The scale error and friction shall not exceed the tolerances specified in Tables I and II for low temperature. The indicator shall then be maintained at room temperature for a period of not less than four hours



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and the indicator shall again be subjected to the Scale error and friction test. The room temperature tolerances of Tables I and II apply. The indicator shall be examined externally for deterioration or damage due to extreme temperature exposure No damage or deterioration shall have occurred which will affect subsequent operation.

4 6 16 High temperature exposure - The indicator shall be tested in accordance with Procedure II of MIL-E-5272 except that the exposure period shall be 24 hours The friction and scale errors at the high temperature shall not exceed the tolerances specified in Tables I and II for high temperature The indicator shall be returned to room temperature for a minimum period of four hours and shall then be subjected to, and meet the requirements of 4 6 4

4 6 17 Thermal shock - The indicator shall be subjected to eight cycles of exposure to water at 85 ±2° C (185 ±3 6° F) and 5 ±2° C (41 ±3 6° F) without evidence of moisture penetration or damage to coating or enclosure. Each cycle of the test shall consist of the indicator being immersed in water at 85° C for a period of 25 minutes, and then being immersed for a period of 25 minutes in the 5° C water bath The time between baths shall not exceed 5 seconds The cycles shall be continuous, one cycle following the other, until eight cycles have been completed The indicator shall then be subjected to the Sealing test No indicator leakage shall occur as a result of this test

4 6 18 Fogging - The indicator shall be energized and placed in a 71° ±2° C controlled ambient temperature for a minimum period of one hour After this period and while still at this temperature, an ice cube shall be rubbed on the cover glass for a period of from one to two minutes The glass shall be wiped dry (do not use compressed air) and the indicator inspected for evidence of water or oil fog inside the cover glass Evidence of fogging shall be cause for rejection

4 6 19 Radio noise interference - The electrical system of the indicator shall be tested for conducted and radiated radio noise interference in accordance with MIL-I-6181 The indicator should be tested under the following conditions (1) Energized at any indication selected by the inspector (2) The same as (1) except that random vibration shall be applied such as lightly tapping or shaking of the indicator (3) Energized and cycling from 10 to 80 percent full scale and return at a rate of 12 ±1 cycles per hour The indicator case shall be grounded by connecting pin "J" to the low side of the power source, unless otherwise specified by the procuring activity

4 6 20 Life - The indicator shall be subjected to operation from 10 percent to approximately 80 percent of end scale and return 12 ±1 times each hour on a continuous basis except that there shall be a power shutdown of 5 minutes in each hour At the conclusion of 1000 hours total time the indicator shall be subjected to and meet the requirements of 4 6 4 and 4 6 6 For lighted type indicators the Life test of MIL-L-25467 shall be conducted simultaneously with the Life test specified herein.



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4.6.21 Salt spray - The indicator shall be tested in accordance with Procedure I of MIL-E-5272. The indicator shall then be examined and there shall be no corrosion or deterioration which will affect subsequent operation. The external surface of the indicator shall then be washed with distilled water and air-dried. The indicator shall be subjected to and meet the requirements of the Dielectric and Sealing tests.

4.6.22 Internal examination - The case of the indicator shall be opened and the mechanism of the indicator shall be examined. The lamp circuit of the lighted type indicator also shall be checked as specified in MIL-L-25467. Any deterioration or damage which could in any manner prevent the indicator from meeting functional maintenance or service requirements during service life shall be cause for rejection.

4.6.23 Lighting contrast - The lighted type indicator shall be subjected to, and meet the requirements of the Contrast test specified in MIL-L-25467.

4.6.24 Mechanical characteristics - The three secondary leads of the indicator synchro shall be connected to the corresponding leads of a transmitter synchro conforming to MIL-T-5350. A normal excitation of 26 volts, 400 cycles shall be supplied to the primaries of both synchros. The rotor of the transmitter synchro shall be locked in one position and the rotor of the indicator synchro rotated through an angle from the corresponding synchronous position. The torque tending to return the rotor of the indicator synchro to its synchronous position shall be measured. The torque is proportional to the angular displacement within displacements up to 20 degrees and shall be no less than 8,000 milligram millimeters per degree of displacement.

5. PREPARATION FOR DELIVERY

5.1 Packaging - The indicator shall be packaged in accordance with MIL-STD-794, Level A or C as specified in contract or order (see 6.2). For Level A packaging the method of preservation shall be in accordance with MIL-P-116, Method IC, without preservative compound. The container shall be reusable.

5.2 Packing - The indicator shall be packed in accordance with MIL-STD-794 Level A, B, or C as specified in contract or order (see 6.2).

5.3 Marking - The interior and exterior containers shall be marked as specified in MIL-STD-794.

5.3.1 Precautionary marking - The following precautionary markings shall appear on two opposite sides of each interior package and shipping container wherever practicable.

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE



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## 6. NOTES

6.1 Intended use - The indicator covered by this specification is intended for use in aircraft to indicate, remotely, the rate of fuel flow.

6.2 Ordering data - Procurement documents should specify the following

(a) Title number and date of this specification.

(b) The quantity, MS part number and type number if any, of the instrument desired.

(c) Levels of packaging (see 5.1) and packing (see 5.2) desired.

(d) The laboratory that shall conduct tests (see 4.4, 4.4.2.2)

(e) Reliability test phases (see 4.3)

6.3 Definitions -

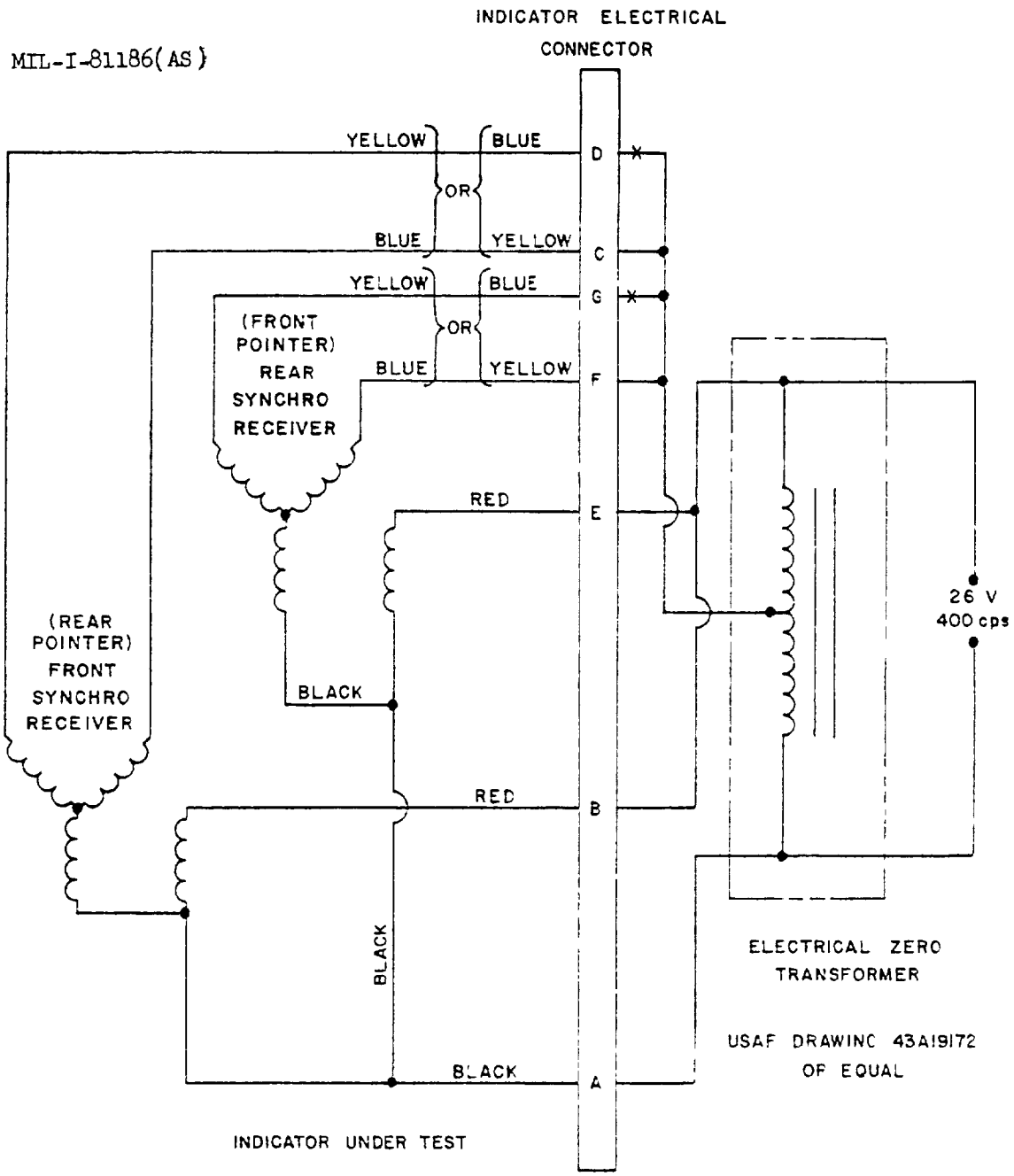
6.3.1 Amplitude - Whenever the word "amplitude" is used, it shall mean the extent of motion as measured from one extreme to the opposite extreme

6.3.2 Hermetic seal - A hermetic seal shall be a perfectly closed and air tight seal made between metallic and metallic or between metallic and vitric materials. A hermetic seal is not intended to include seals which are dependent upon gaskets

6.3.3 Electrical zero - With the indicator connected and energized as shown in Figure 4 the tip of the pointer(s) shall be within 0.010 inch of the electrical zero reference mark on the scale. For dual indicators the



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NOTE

DUAL INDICATOR SHOWN FOR SINGLE INDICATOR OMIT PINS G, F, AND E

Figure 4. Electrical Zero Test Connections



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pointer tips shall also be within 0.010 inch of each other. Disconnection at pin D, and G for dual indicators, shall cause the pointer(s) to move approximately 30 electrical degrees in the increasing scale direction.

6.3.4 Good as new - Good as new" shall mean indicators operated less than 10 percent of the specified MTBF Operation.

6.3.5 Refurbished - Refurbished shall mean that the indicator has been completely overhauled with all component parts meeting current parts standards and the indicator shall have been subjected to and met all the requirements of a new indicator.

6.4 Qualification - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for the opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement and manufacturers are urged to arrange to have the product that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Department of the Navy, Washington, D.C. 20361 and information pertaining to qualification of products may be obtained from that activity.

6.5 Precedence of documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict the following precedence shall apply.

(a) Contract - The contract shall have precedence over any specification.

(b) This specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(c) Referenced specifications - The referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines or reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Submitter) | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426

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Naval Air Systems Command
Washington, D. C. 20360

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